```
    d3.selectAll("path.graticule").attr("d", geoPath);
      d3.selectAll("path.countries").attr("d", geoPath);

    d3.selectAll("circle.cities")
        .attr("cx", function(d) {return projection([d.x,d.y])[0]})
        .attr("cy", function(d) {return projection([d.x,d.y])[1]});
};
```

Also calls the now-updated projection

Any path will be properly redrawn by calling the d3.geo.path associated with the updated projection.

The zoom behavior updates its `.translate()` array in reference to your dragging behavior, and increases or decreases the `.scale()` value in reference to your mousewheel and double-click behavior. Because it's designed to work with SVG transform and D3 geographic projections, `d3.behavior.zoom` is all you need for pan-and-zoom functionality.

Infoviz term: semantic zoom

When you think about zooming in on things, you naturally think about increasing their size. But from working with mapping, you know that you don't just increase the size or resolution as you zoom in; you also change the kind of data that you present to the reader. This is known as *semantic zoom* in contrast to *graphical zoom*. It's most clear when you look at a zoomed-out map and see only country boundaries and a few major cities, but as you zoom in you see roads, smaller cities, parks, and so on.

You should try to use semantic zoom whenever you're letting your user zoom in and out of any data visualization, not just a chart. It allows you to present strategic or global information when zoomed out, and high-resolution data when zoomed in.

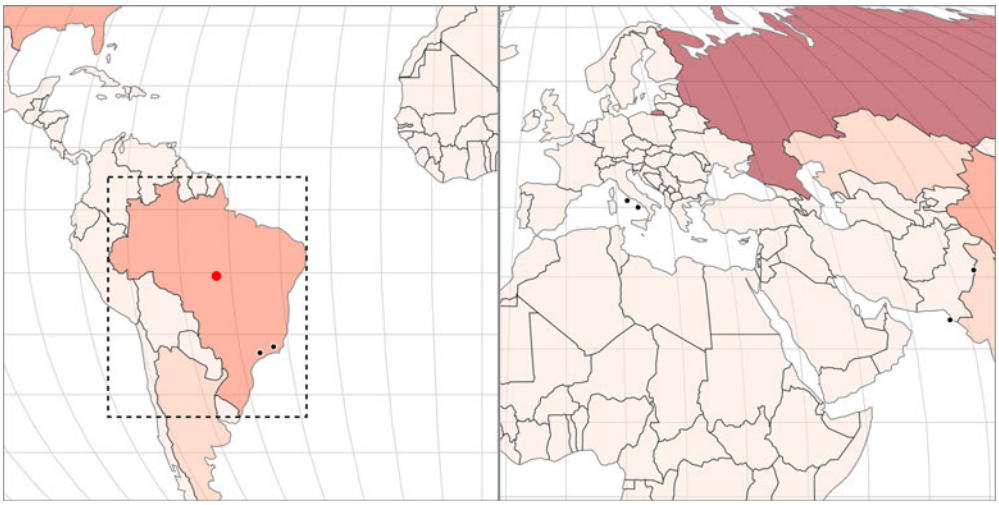


Figure 7.9 Our map with zooming enabled. Panning occurs with the drag behavior and zooming with mousewheel and/or double-clicking. Notice that the bounding box and centroid functions still work, because they're based on our constantly updating projection.